Planar Array of Microstrip Patch Antennas

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Background

- Select 5.8 GHz ISM band
  - Spans 150 MHz from 5.725 to 5.875 GHz
- Compared to 2.4 GHz ISM band, it offers:
  - 50 MHz more bandwidth (2.4 GHz allocates 100 MHz from 2.4 to 2.5 GHz)
  - Lower in-band interference: ‘less crowded’
- An array of these elements is suitable for long-range video applications
- Design requirements summarized:
  - f_R = 5.8 GHz
  - G_{PK} = 20 dBi
  - %BW > 3%
  - E to E spacing >0.5λ

Element Design

- Analytical equations provide a starting point:
  - $W = \frac{c}{2f_R\sqrt{\epsilon_{avg}}} = 20.4$ mm
  - $L_R = \frac{c}{2f_R\sqrt{\epsilon_{re}}} - 2\Delta L = (18.136 - 0.832)$ mm = 16.471 mm
  - $y_{off} = \frac{L_R}{\pi}\cos^{-1}\left(\sqrt{\frac{Z_0}{Z_{edge}}}\right) = 6.098$ mm
- However, implemented values were modified to:
  - $W = 19.707$ mm
  - $L_R = 15.746$ mm
  - $y_{off} = 4.398$ mm

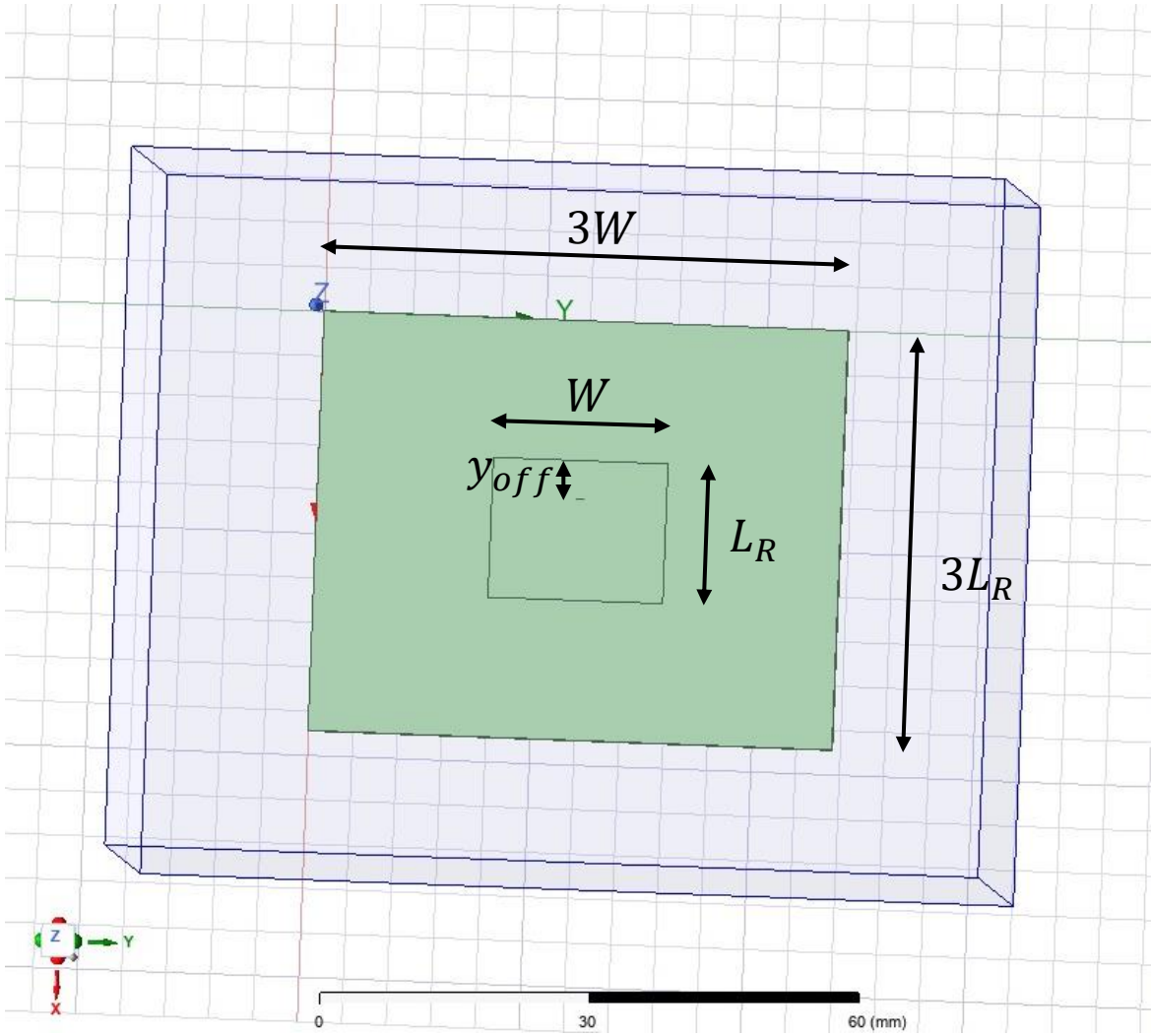


Fig. 1. Microstrip patch antenna simulation model

- f_R = 5.77 GHz
- %BW = $\frac{5.8825-5.6605}{0.0577}$ = 3.85%

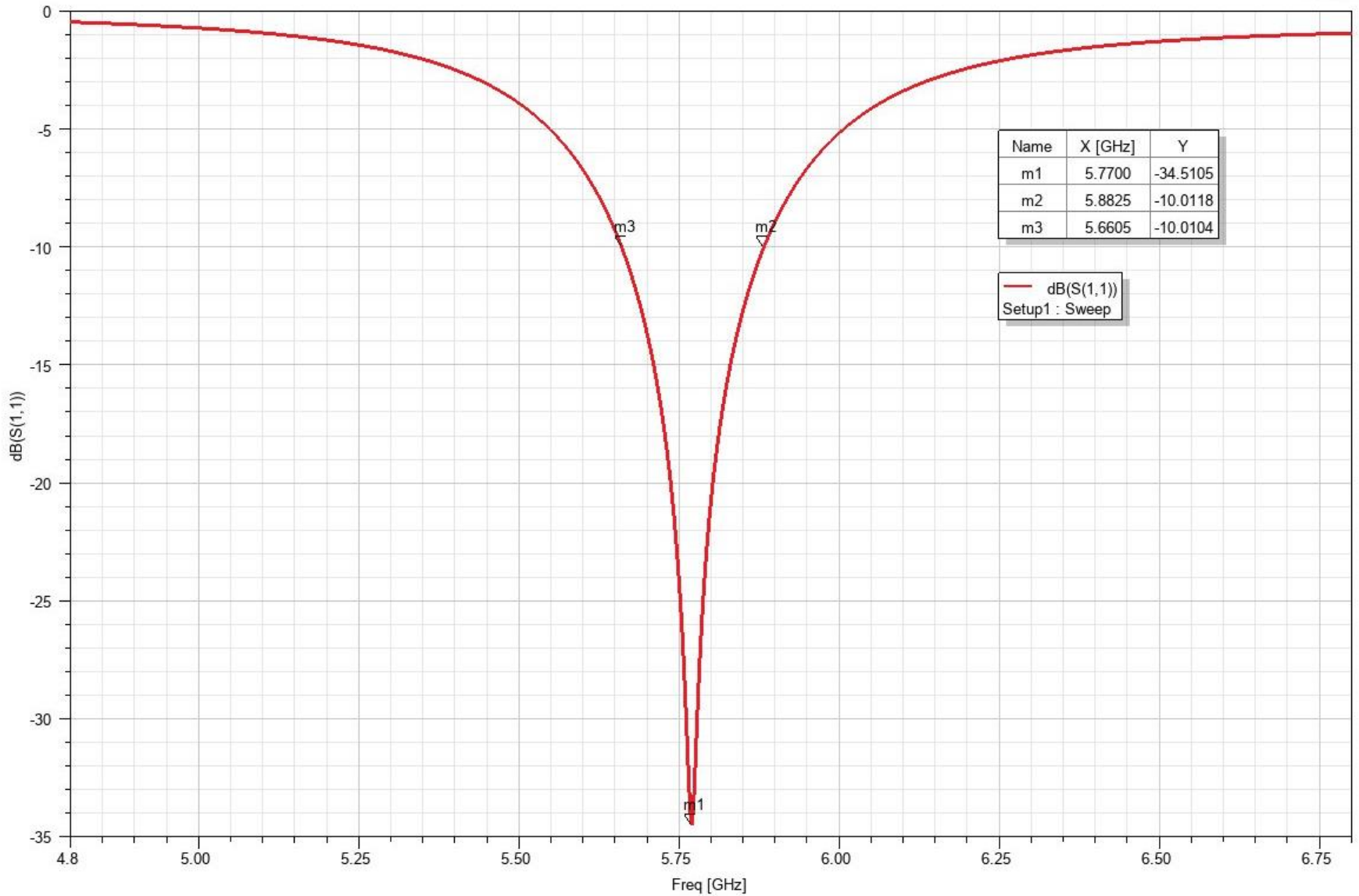


Fig. 2. Element rectangular plot of S(1, 1)

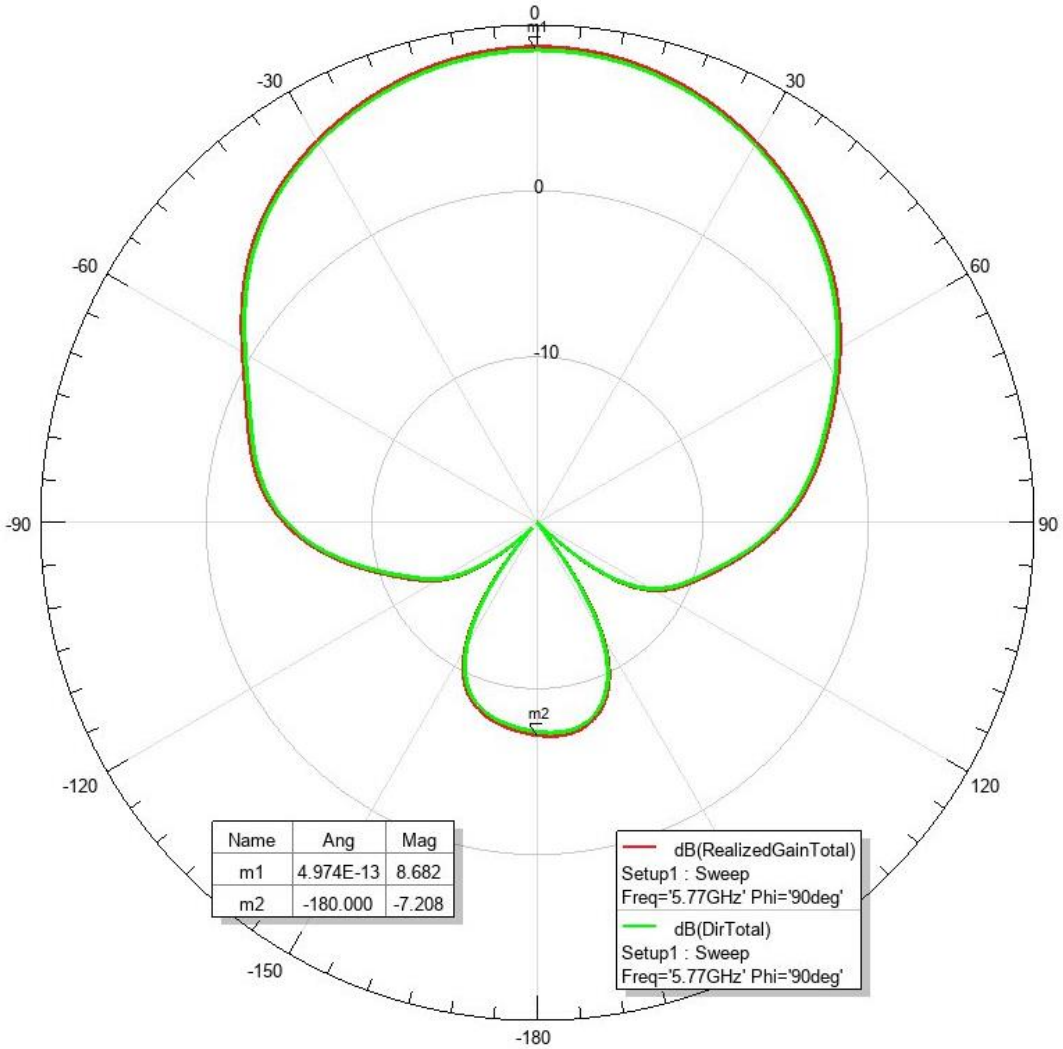


Fig. 3. Element ϕ = 90° radiation pattern

- G_{PK} = 8.7 dBi
- F/B = 15.9 dB

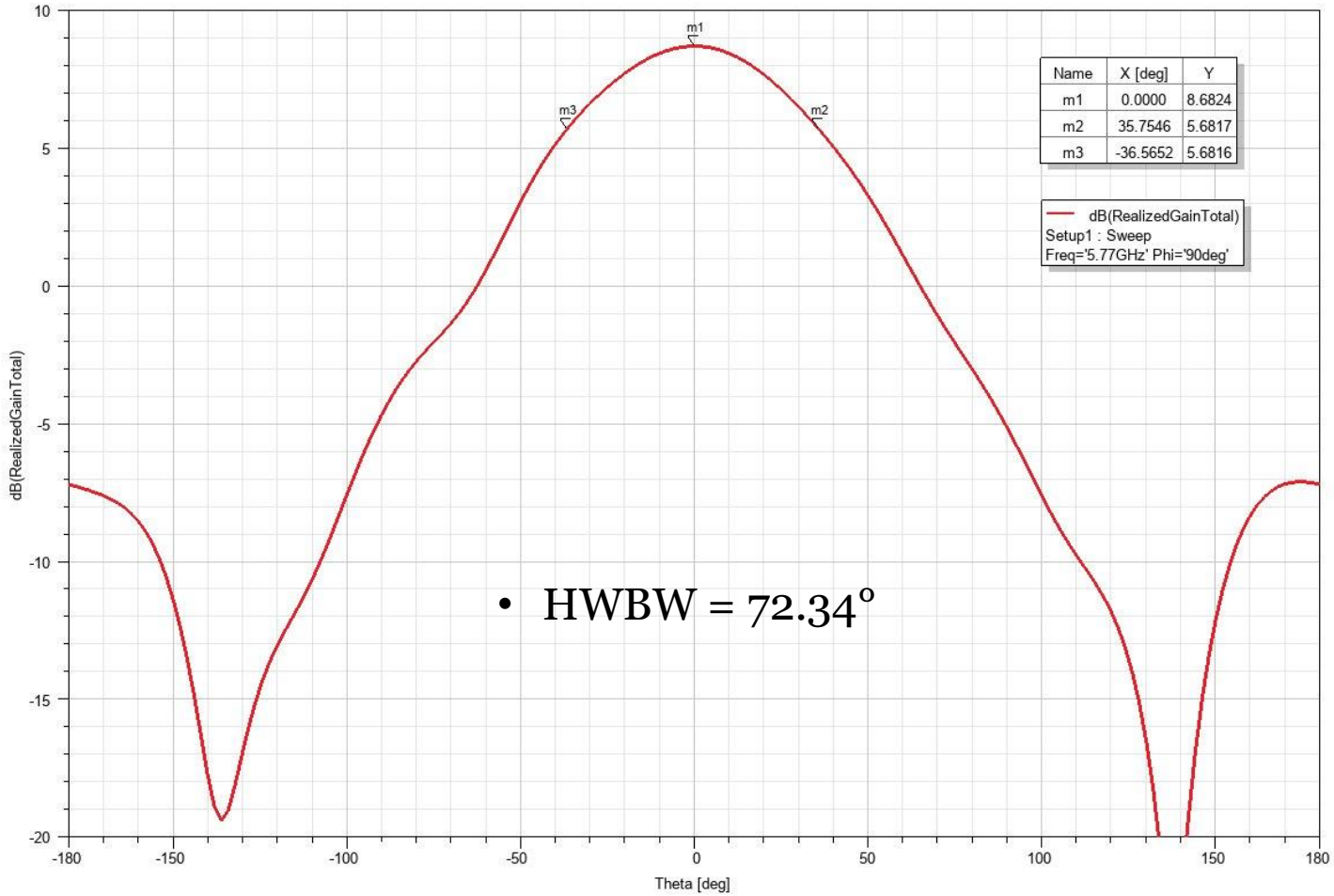


Fig. 4. Element ϕ = 90° rectangular radiation pattern

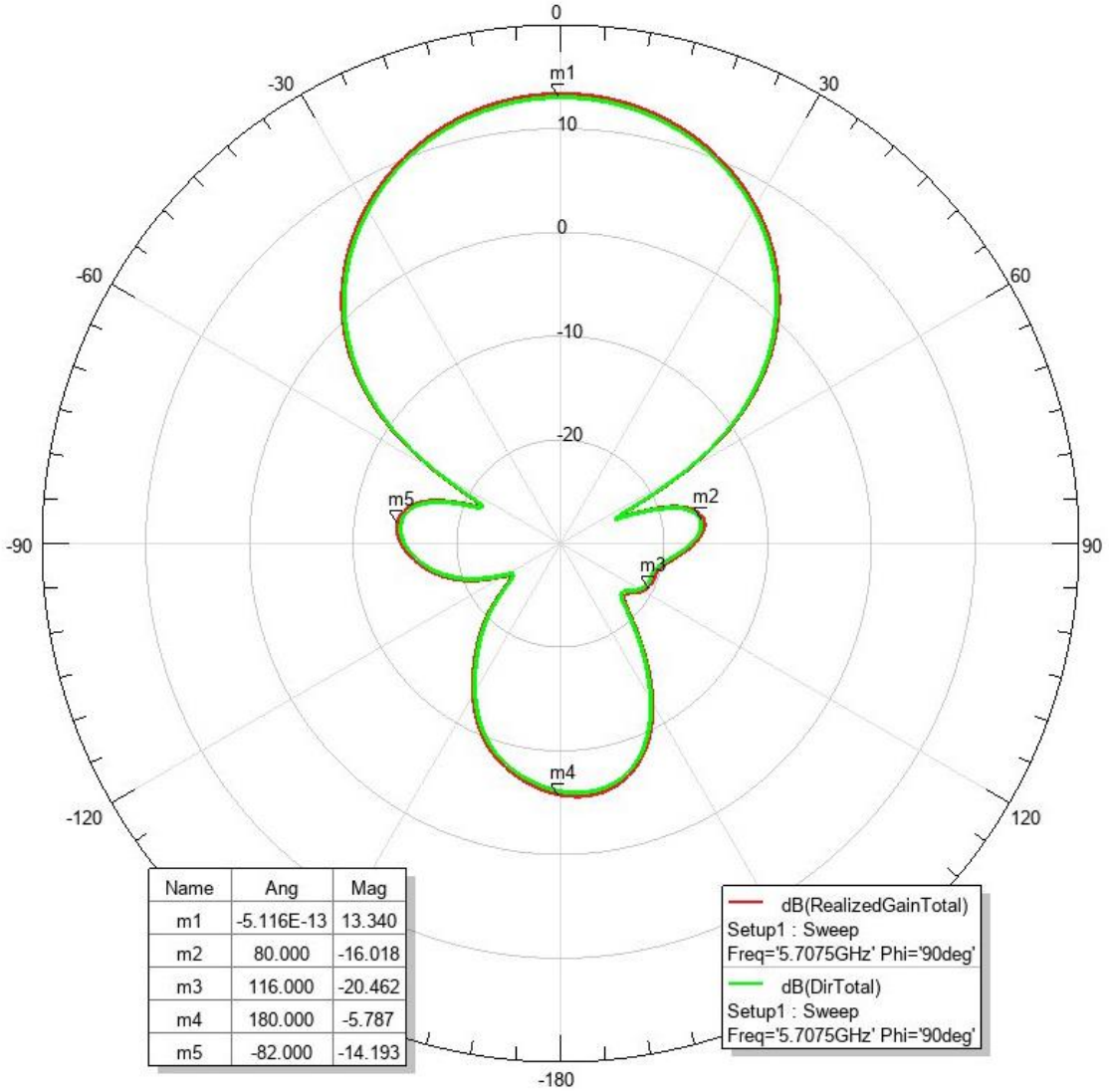


Fig. 5. 2×2 array radiation pattern ϕ = 90°

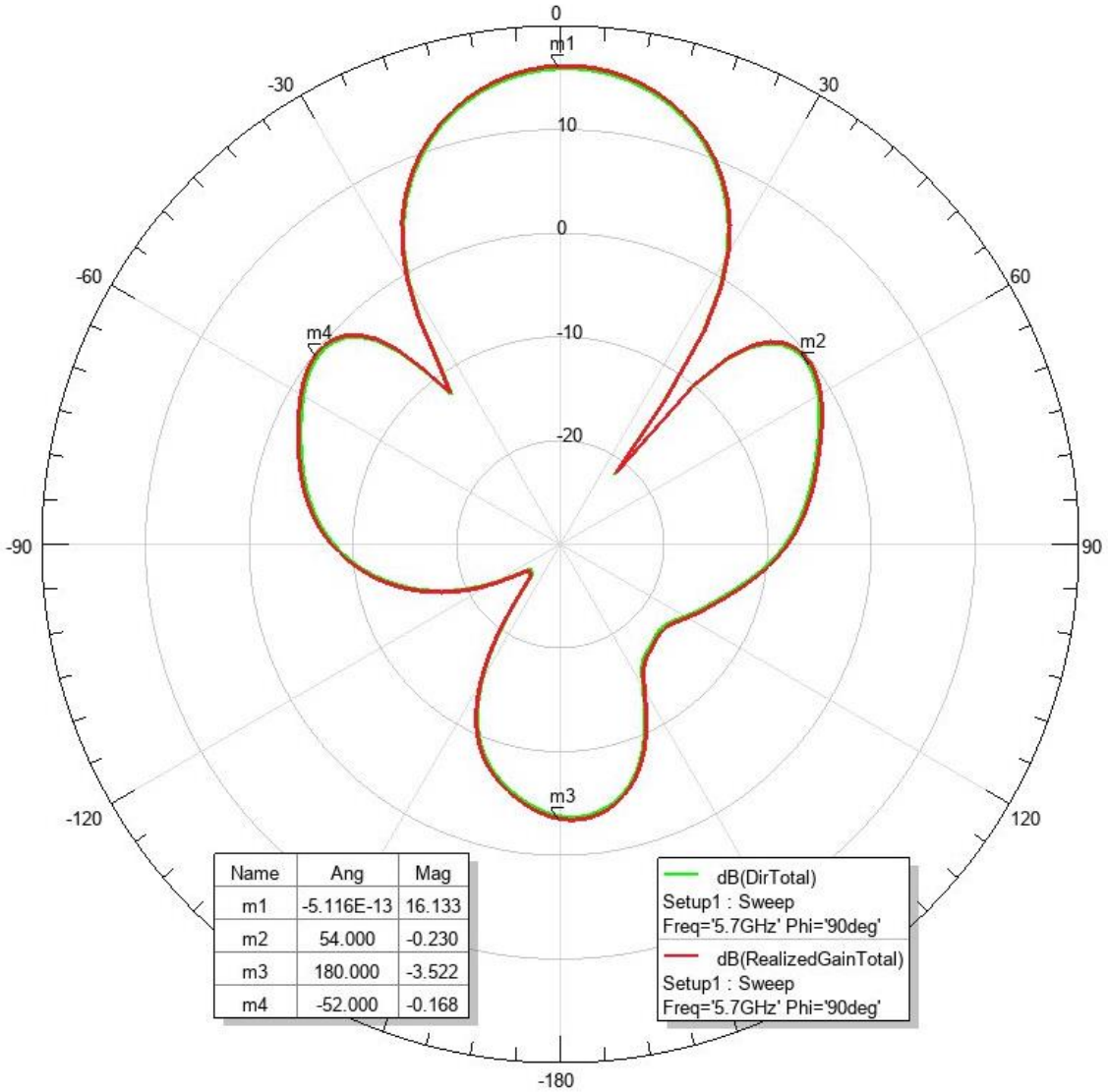


Fig. 6. 3×3 array radiation pattern ϕ = 90°

5×5 Planar Array Design

- Element to element spacing selected l = $0.5625 \cdot \lambda = 29.348$ mm
  - For $\lambda = 52.174$ mm at 5.75 GHz
- Stored l in project variables to greatly aid model revisions

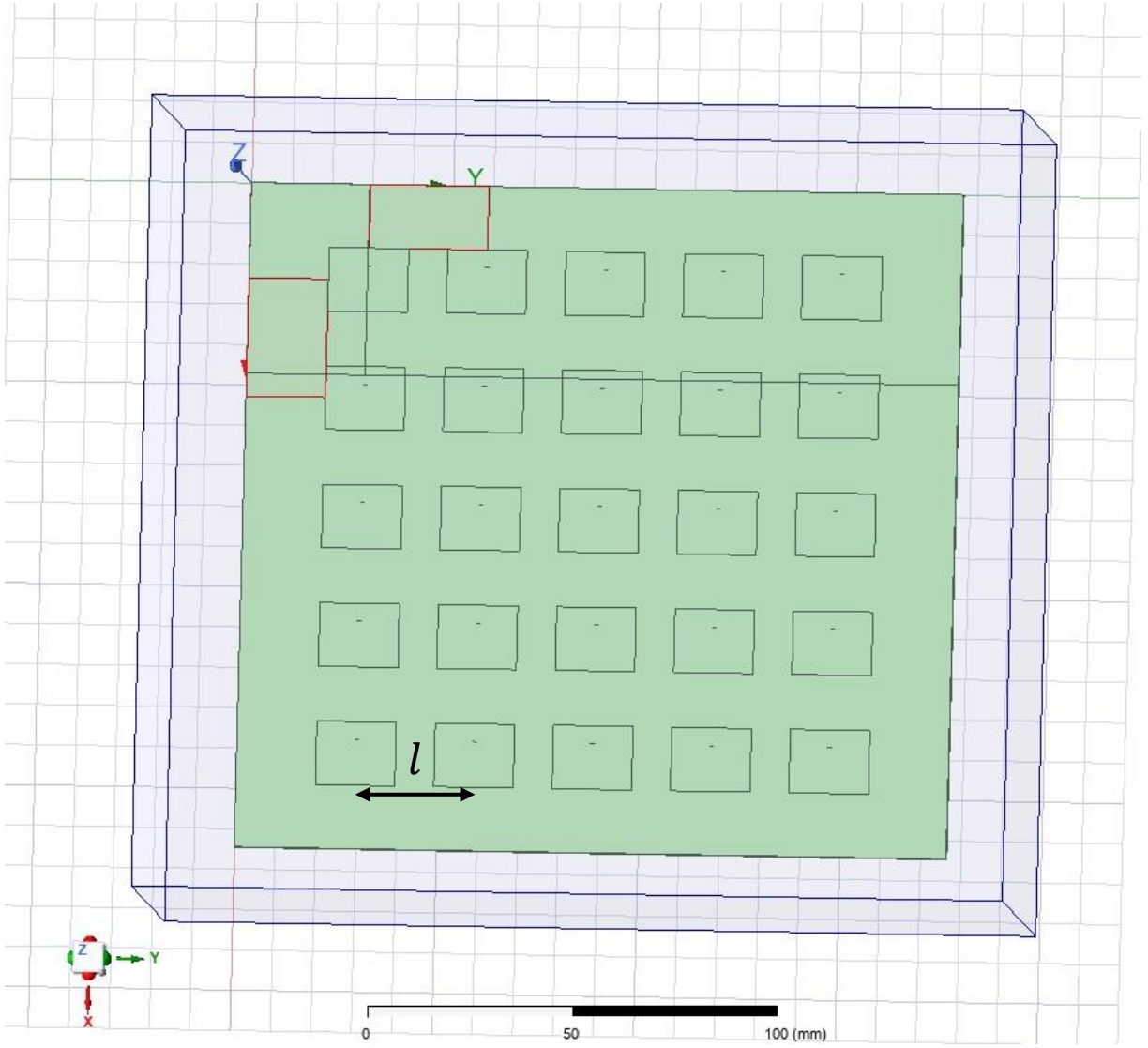


Fig. 7. 5×5 planar array of microstrip patch elements

- $f_R = 5.645$ GHz
- $\%BW = \frac{5.7525-5.5450}{0.05645} = 3.68\%$

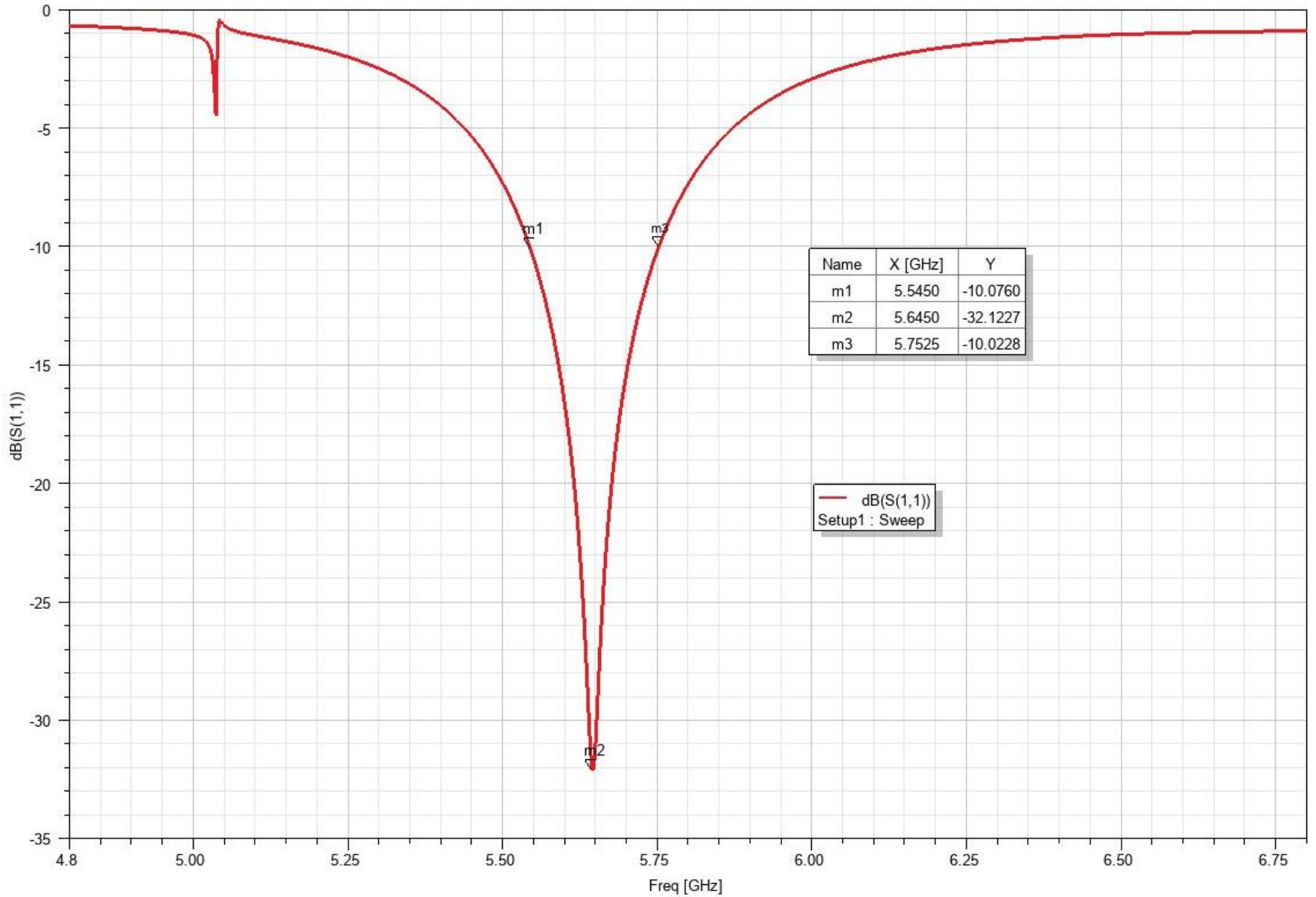


Fig. 8. Array plot of S(1, 1)

- At $f = f_R$
  - $|s_{21}|$ = -18.8 dB
  - $|s_{23}|$ = -22.4 dB

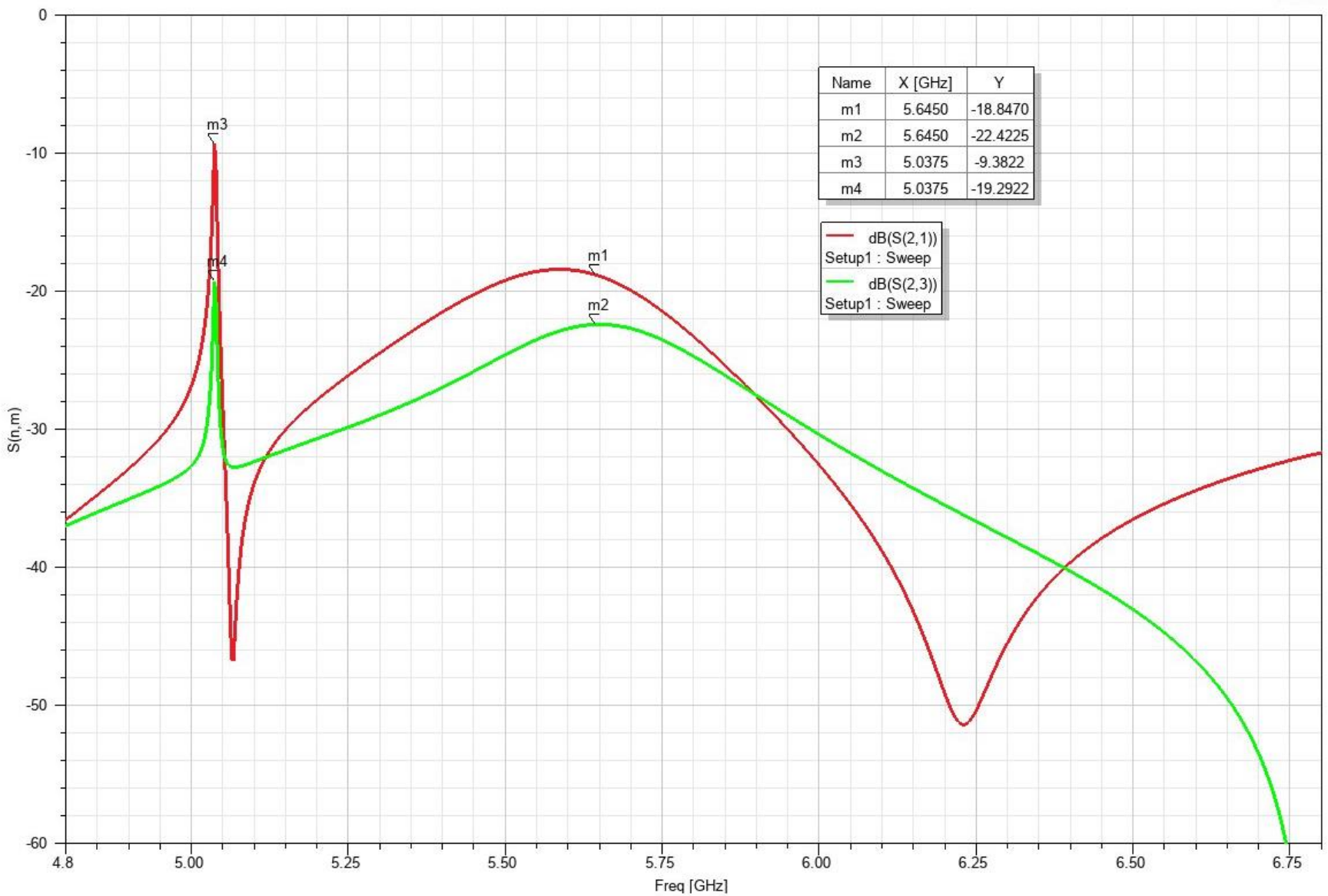
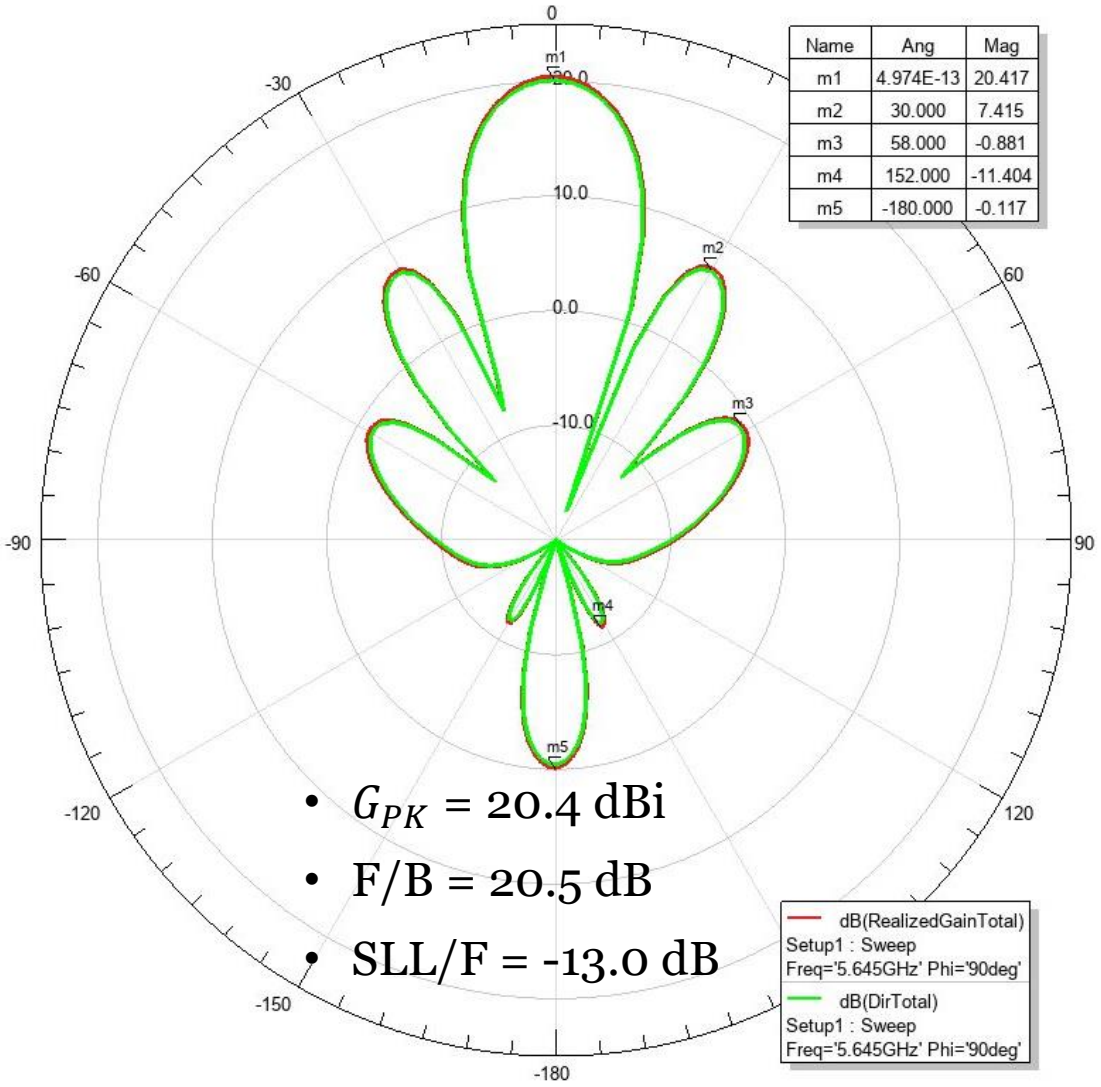
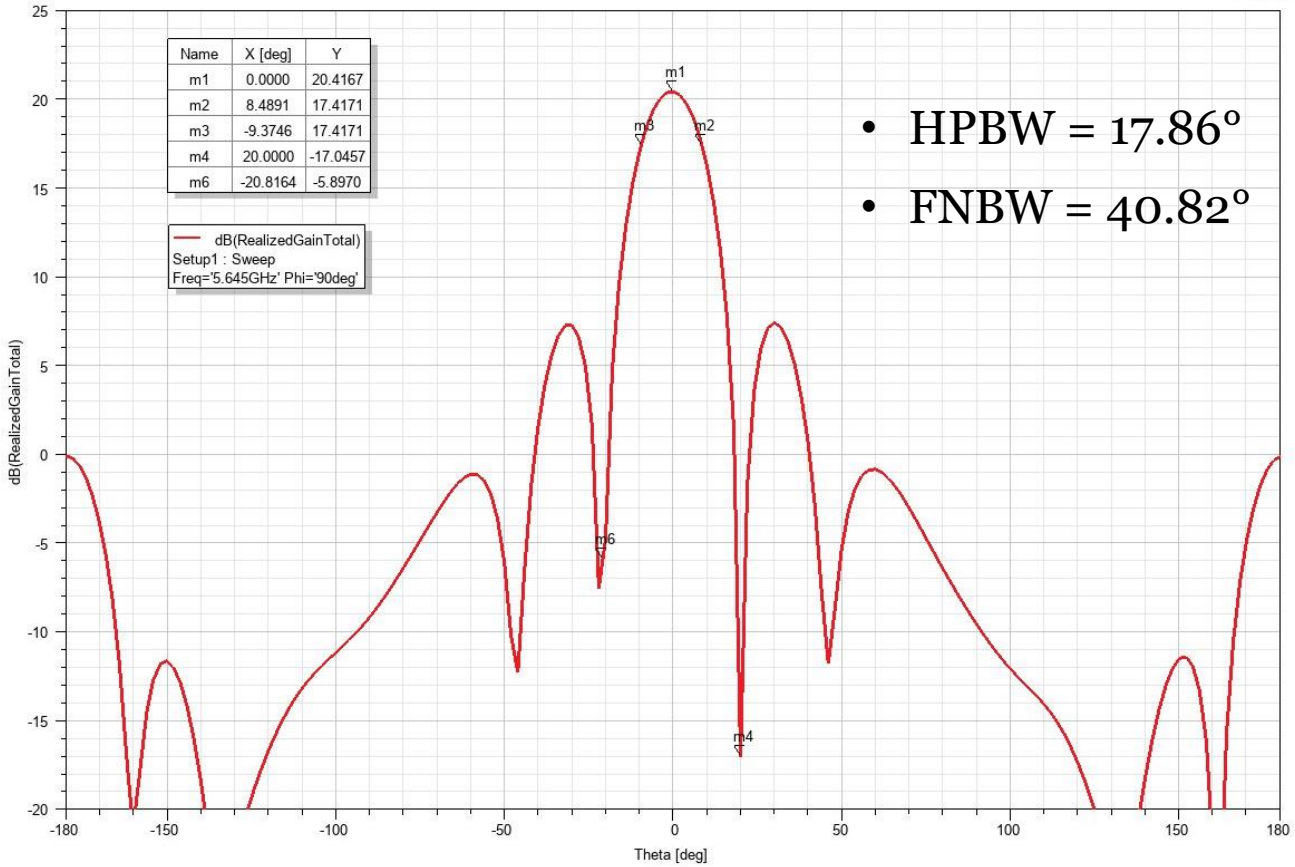


Fig. 9. Array plot of S(2, 1) and S(2, 3)



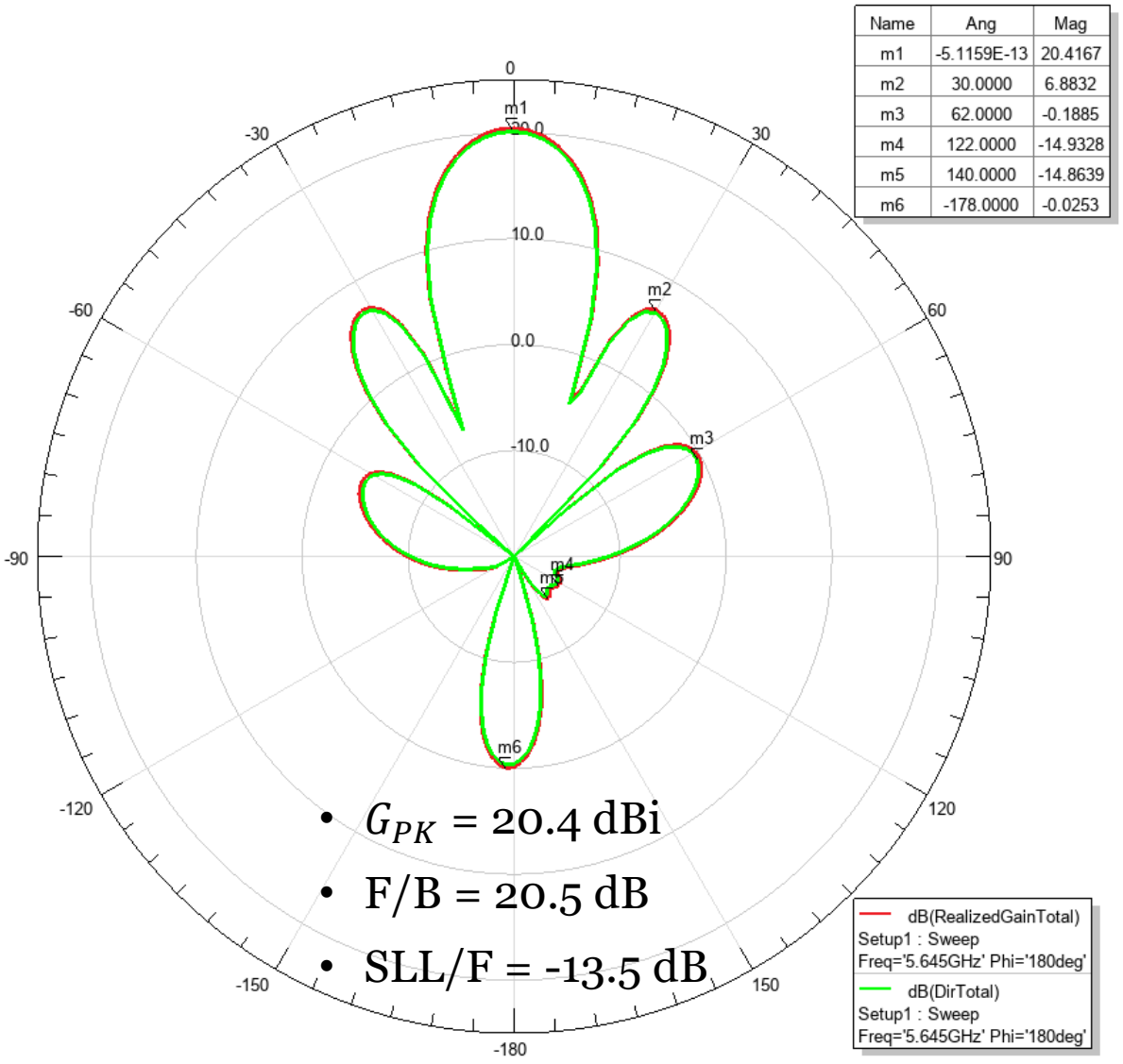
- G_{PK} = 20.4 dBi
- F/B = 20.5 dB
- SLL/F = -13.0 dB

Fig. 10. Array ϕ = 90° radiation pattern



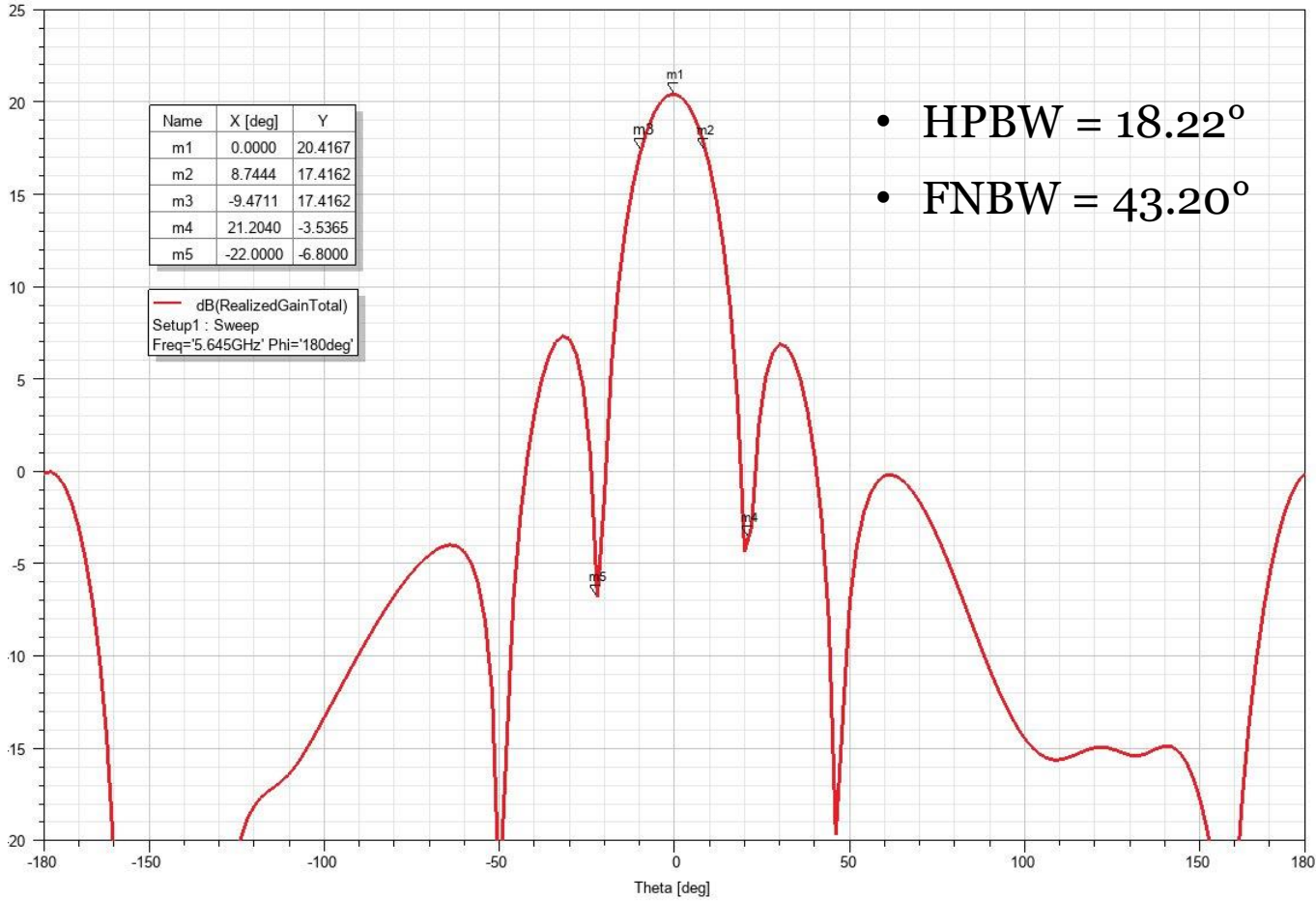
- HPBW = 17.86°
- FNBW = 40.82°

Fig. 11. Array ϕ = 90° rectangular radiation pattern



- G_{PK} = 20.4 dBi
- F/B = 20.5 dB
- SLL/F = -13.5 dB

Fig. 12. Array ϕ = 180° radiation pattern



- HPBW = 18.22°
- FNBW = 43.20°

Fig. 13. Array ϕ = 180° rectangular radiation pattern

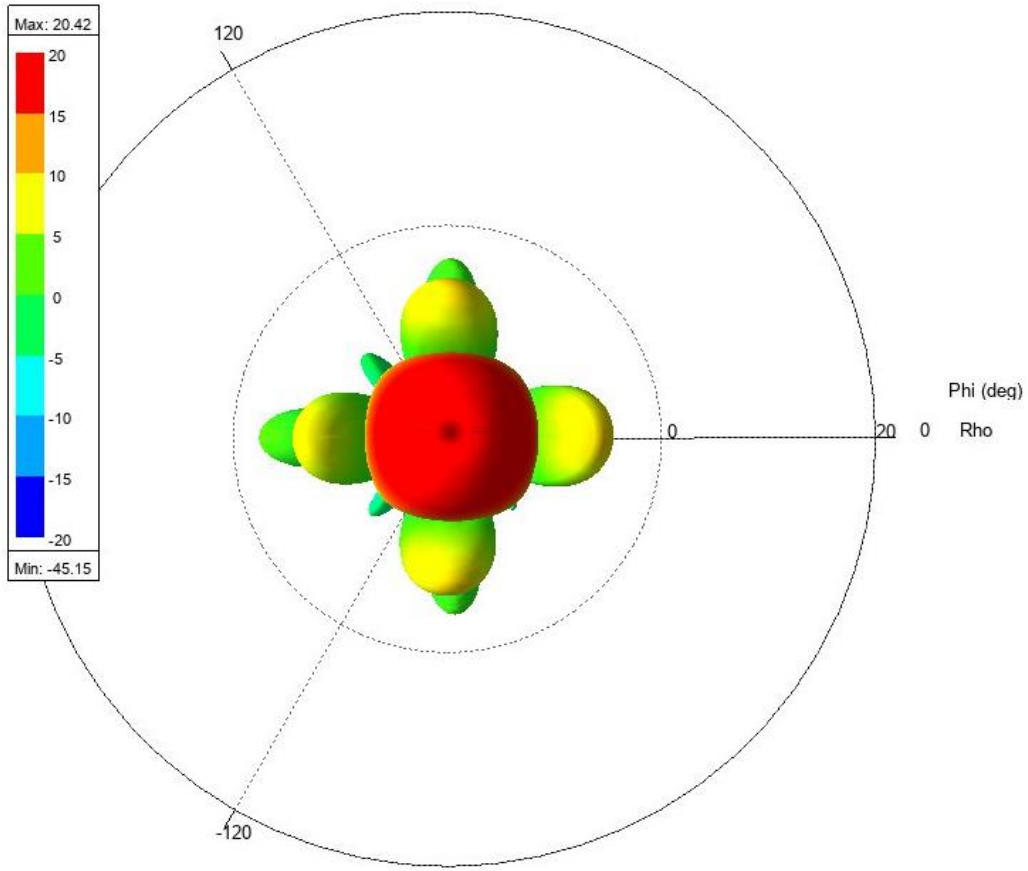


Fig. 14. Array 3D radiation pattern (top)

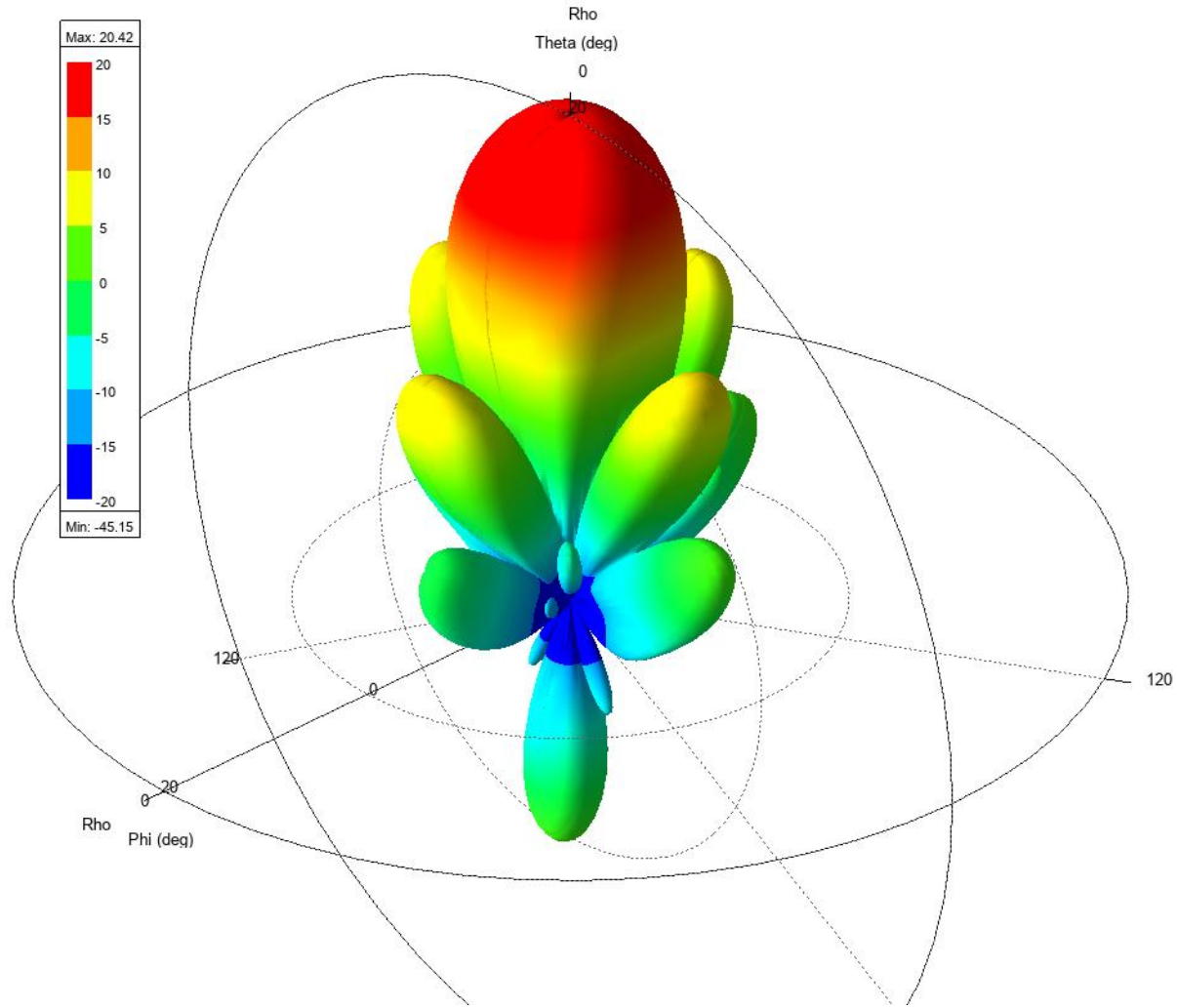
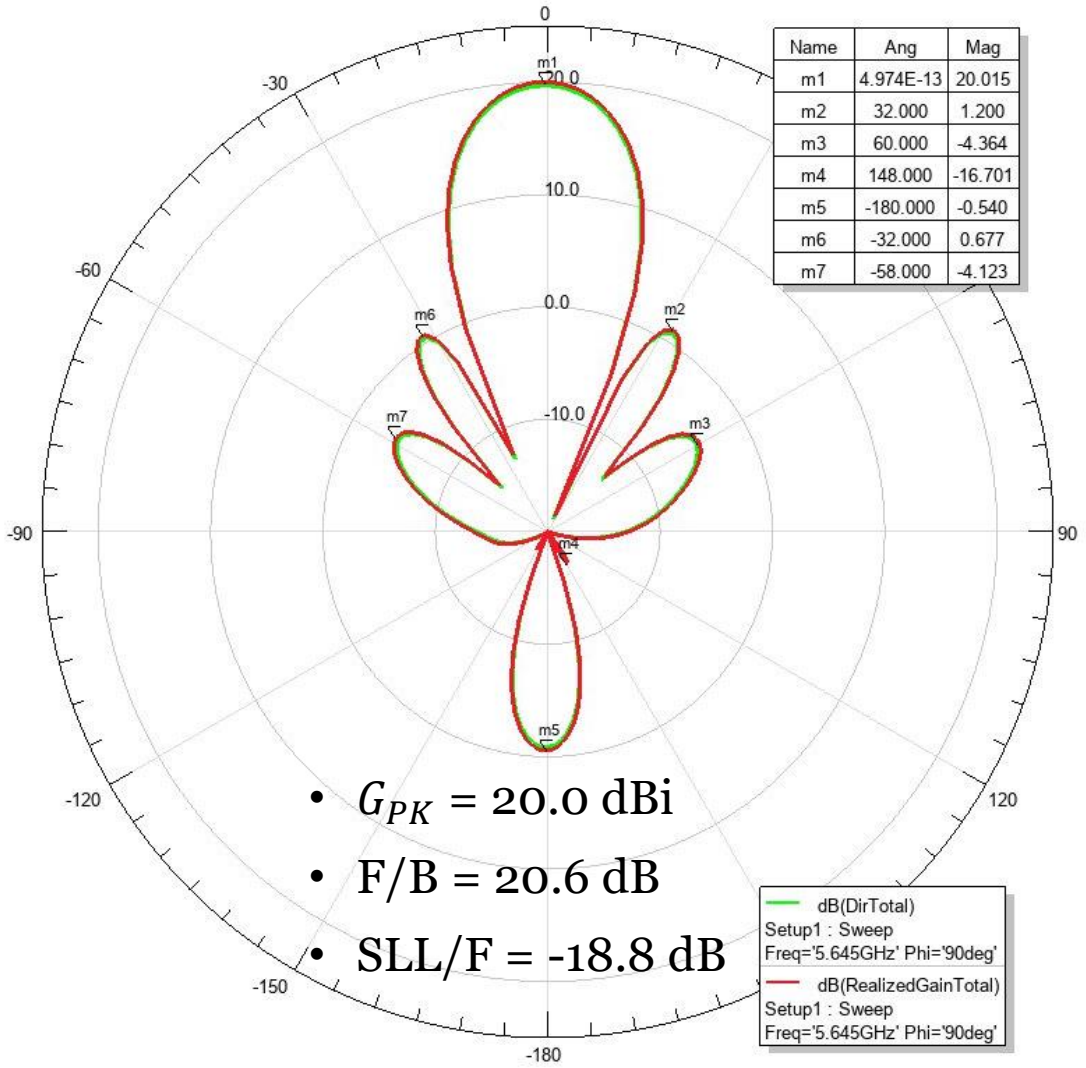
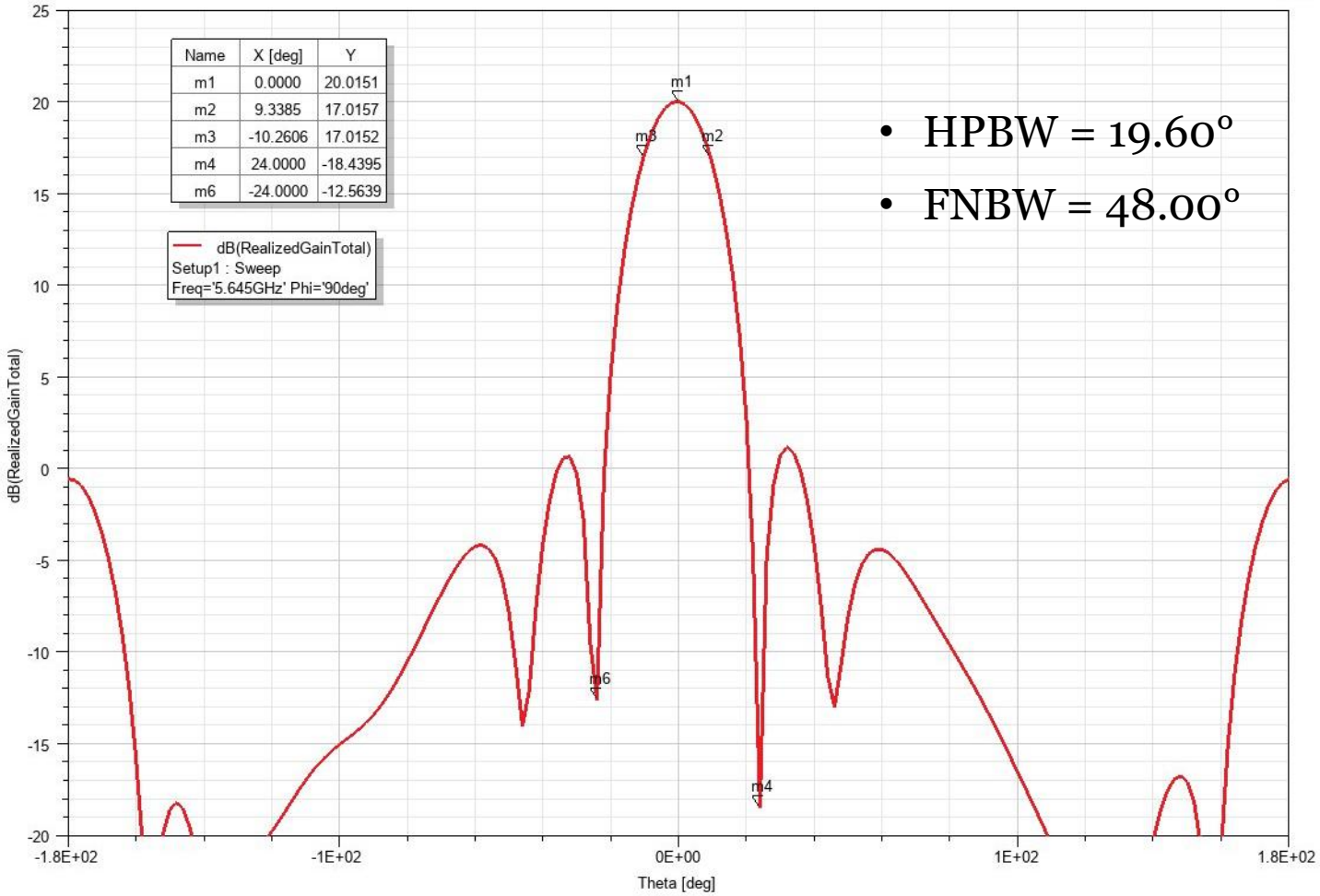


Fig. 15. Array 3D radiation pattern (side)



- G_{PK} = 20.0 dBi
- F/B = 20.6 dB
- SLL/F = -18.8 dB

Fig. 16. Array ϕ = 90° radiation pattern with taper



- HPBW = 19.60°
- FNBW = 48.00°

Fig. 17. Array ϕ = 90° rectangular radiation pattern with taper

- ΔG_{PK}: -2.0%
- ΔF/B: 0.5%
- ΔSLL/F: -44.6%
- ΔHPBW: 9.7%
- ΔFNBW: 11.1%

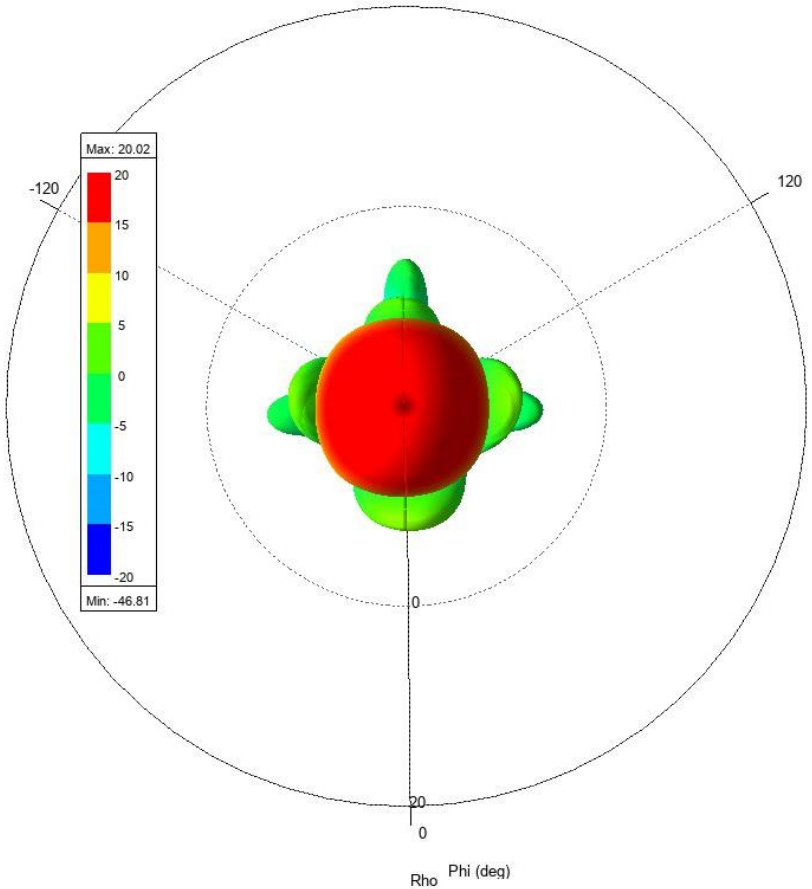


Fig. 18. Array 3D radiation pattern with taper (top)

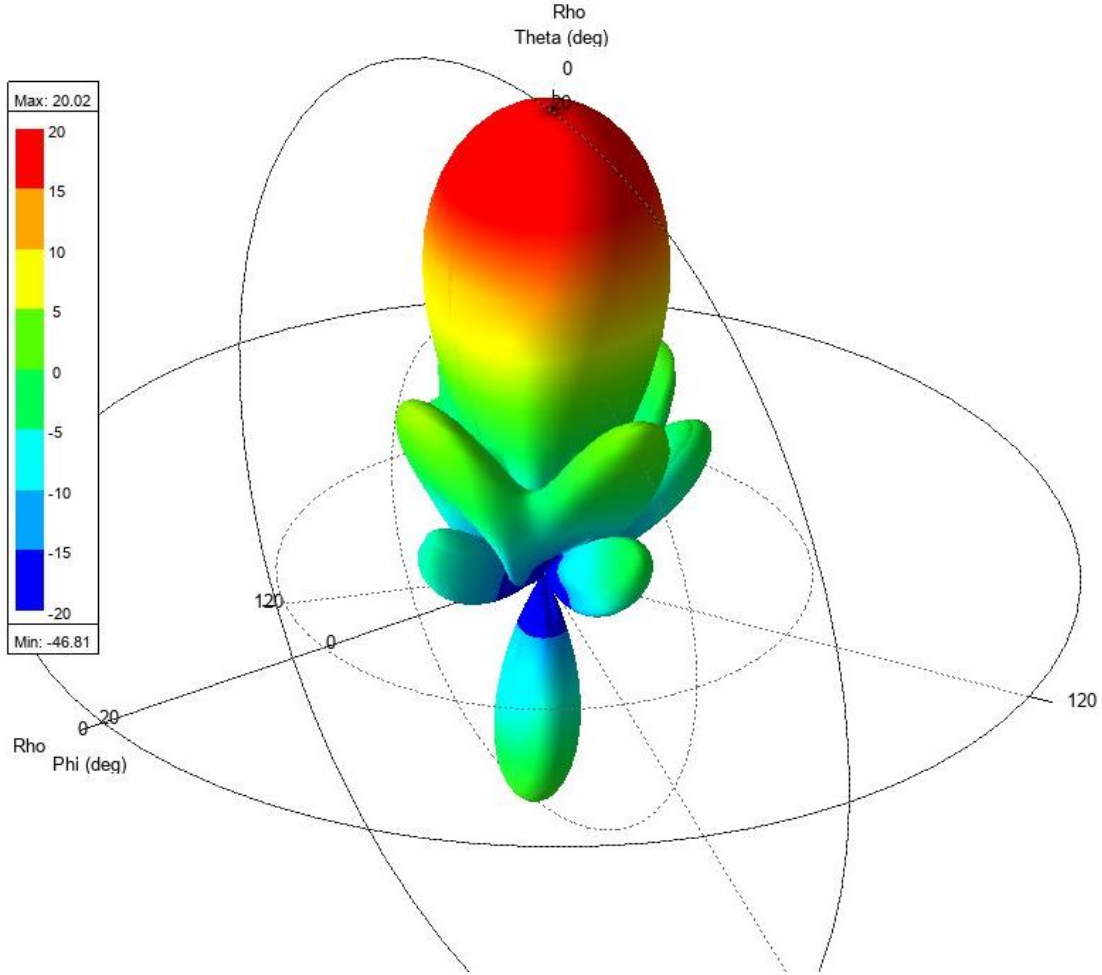


Fig. 19. Array 3D radiation pattern with taper (side)

Closing

- Designer should compensate for mutual coupling of elements to match design frequency
  - f_R less than 5.8 GHz by 2.7%
- Tapered excitations along array can reduce peak SLL with minimal concession of peak gain

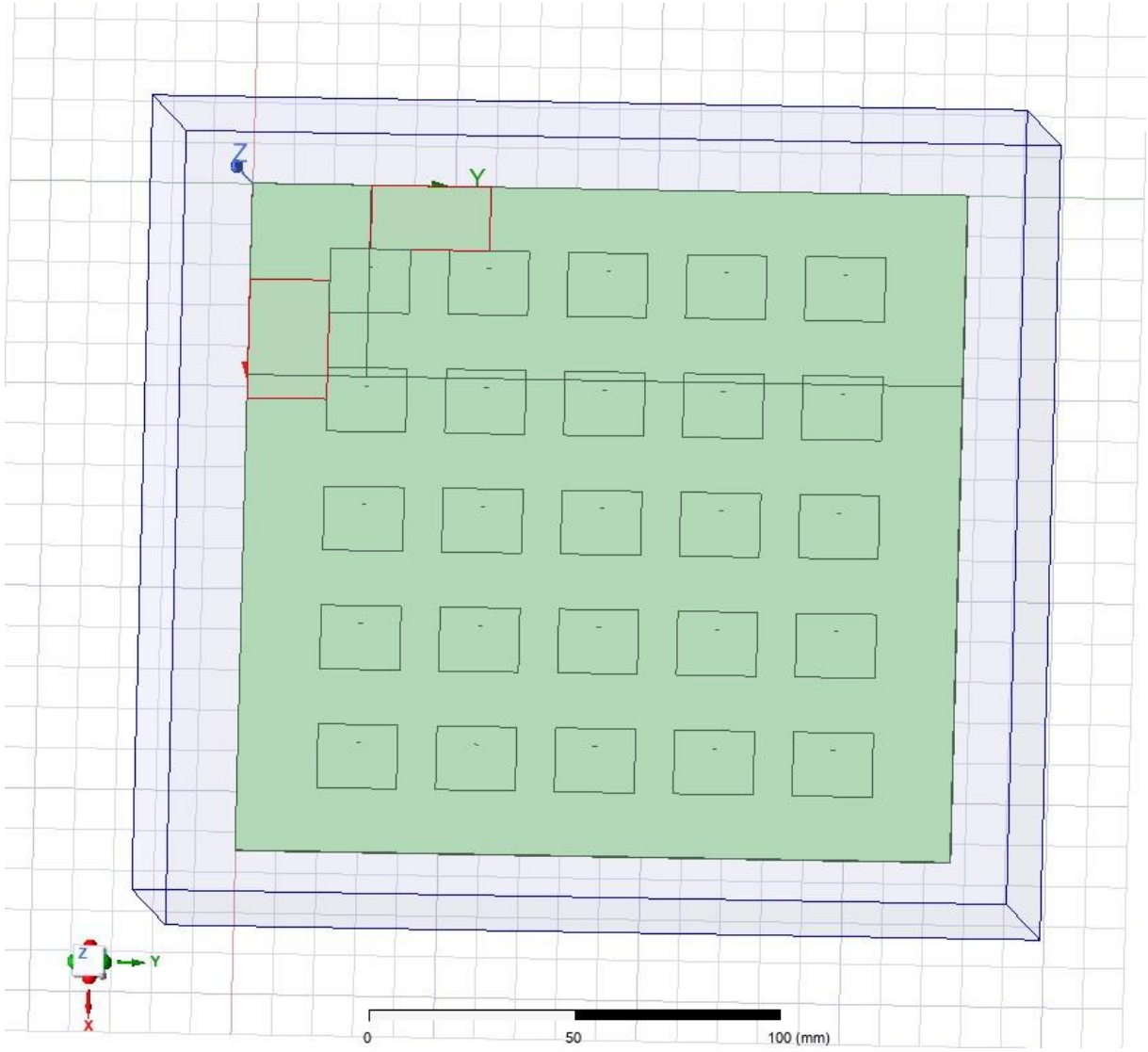


Fig. 6. 5×5 planar array of microstrip patch elements